

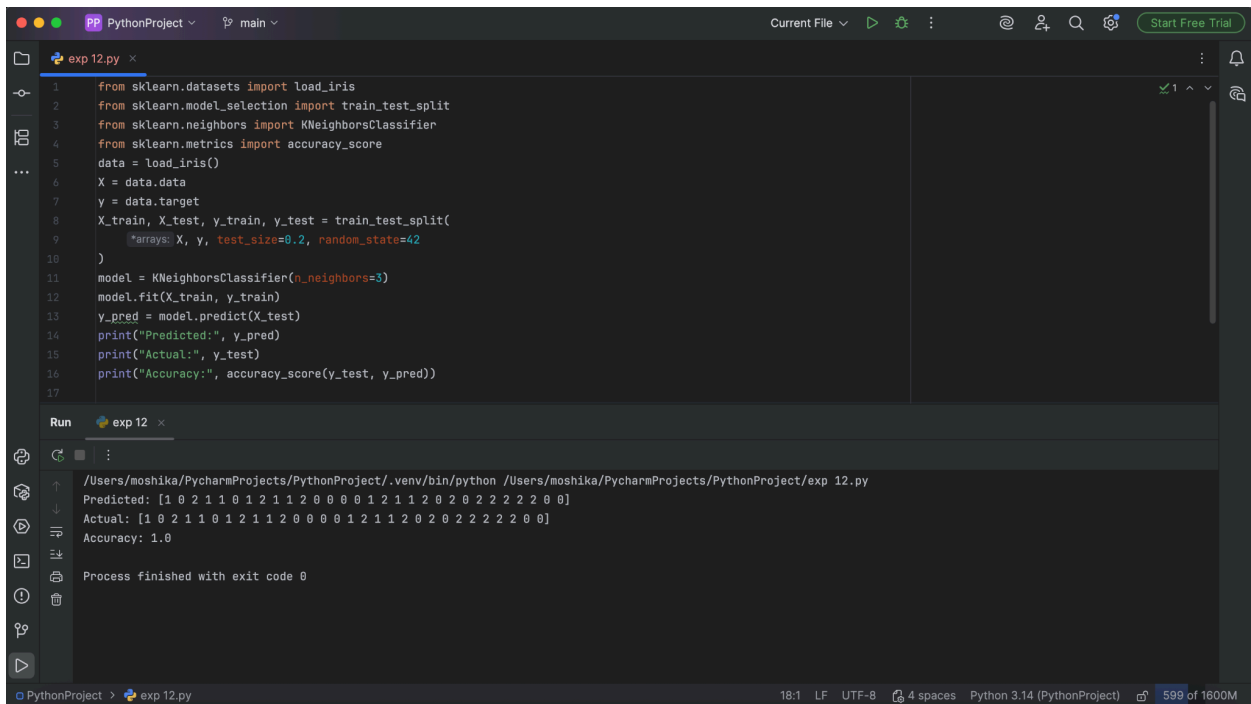
EXPERIMENT - 12

Aim: To classify Iris flowers into different species using the K-Nearest Neighbours algorithm.

Procedure:

1. Load the Iris dataset.
2. Separate features and target labels.
3. Split the data into training and testing sets.
4. Create a KNN classifier.
5. Train the model using training data.
6. Predict the species for test data.
7. Calculate the accuracy.
8. Display the classification results.

Output:



```
1 from sklearn.datasets import load_iris
2 from sklearn.model_selection import train_test_split
3 from sklearn.neighbors import KNeighborsClassifier
4 from sklearn.metrics import accuracy_score
5 data = load_iris()
6 X = data.data
7 y = data.target
8 X_train, X_test, y_train, y_test = train_test_split(
9     *arrays: X, y, test_size=0.2, random_state=42
10 )
11 model = KNeighborsClassifier(n_neighbors=3)
12 model.fit(X_train, y_train)
13 y_pred = model.predict(X_test)
14 print("Predicted:", y_pred)
15 print("Actual:", y_test)
16 print("Accuracy:", accuracy_score(y_test, y_pred))
17
```

Run exp 12

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp_12.py
Predicted: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]
Actual: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]
Accuracy: 1.0
Process finished with exit code 0
```

Result: The KNN algorithm successfully classified Iris flowers into their respective species with good accuracy.